j4_controller -- pin diagram

TTGO T-Display v1.1, rails visible, USB-C at the TOP (as mounted)

```
         LEFT rail                            RIGHT rail
                             +--[ USB-C ]--+
                  3.3V -- 3V3|             |  5V
                ground -- GND|             |  GND  ground
                ground -- GND|     TTGO    |  27   DISP-L RX <- XIAO D6
   (avoid: boot/flash) -- 12 |  T-Display  |  26   DISP-R RX <- XIAO D6
        EYE POP toggle -- 13 |     v1.1    |  25   DISP-R TX (face msgs)
            AUX toggle -- 15 |             |  33   VENT toggle
                (free) -- 2  |   [ ST7789  |  32   LASER toggle
  DISP-L TX -> XIAO D7 -- 17 |    TFT on   |  39   (input only)
               I2C SCL -- 22 |    back ]   |  38   (input only)
               I2C SDA -- 21 |             |  37   (input only)
                ground -- GND|             |  36   (input only)
                ground -- GND|             |  3.3V
                             +-------------+

on-board (no header pin): GPIO34 = battery sense, BOOT = GPIO0,
    GPIO35 = screen-cycle button, TFT on 4/5/16/18/19/23
  I2C bus: ADS_01 @ 0x48, ADS_02 @ 0x49, ADS_03 @ 0x4A, ADS_04 @ 0x4B,
    PCF8574 keypad_left @ 0x21, keypad_right @ 0x20
  toggles: INPUT_PULLUP, switch closes to GND (ON = LOW)
```

GPIO	Function
21	I2C SDA (ADS1115 x4, PCF8574 keypads x2)
22	I2C SCL
17	Serial1 TX to j4_display_left (XIAO D7)
27	Serial1 RX from j4_display_left (XIAO D6)
25	Serial2 TX to j4_display_right (XIAO D7, face-preset messages)
26	Serial2 RX from j4_display_right (XIAO D6, pot feed)
32	LASER toggle (INPUT_PULLUP, switch closes to GND)
33	VENT toggle (INPUT_PULLUP, switch closes to GND)
13	EYE POP toggle (INPUT_PULLUP, switch closes to GND)
15	AUX toggle, next to the right joystick (INPUT_PULLUP, unassigned)
34	Battery voltage sense (ADC input, read-only)
35	Screen-cycle button (TTGO built-in; data / MAC / connection status)

j4_controller -- ADS1115 pin diagrams

pot wiper to the A-pin; outer legs to 3.3V and GND

```
ADS_01 @ 0x48 (ADDR -> GND)                 both joysticks
+------+
| VDD  |  3.3V
| GND  |  GND
| SCL  |  TTGO GPIO 22 (I2C SCL)
| SDA  |  TTGO GPIO 21 (I2C SDA)
| ADDR |  GND  = address 0x48
| ALRT |  not connected
| A0   |  neck joystick X wiper        -> neck L/R differential mix
| A1   |  neck joystick Y (jaw) wiper  -> neck L/R base height
| A2   |  eyes joystick X wiper        -> eyes pan servo  (PCA9685 ch 3), -128..0..128
| A3   |  eyes joystick Y wiper        -> eyes tilt servo (PCA9685 ch 4), -128..0..128
+------+

ADS_02 @ 0x49 (ADDR -> VDD)                 four face pots, left bank
+------+
| VDD  |  3.3V
| GND  |  GND
| SCL  |  TTGO GPIO 22 (I2C SCL)
| SDA  |  TTGO GPIO 21 (I2C SDA)
| ADDR |  VDD  = address 0x49
| ALRT |  not connected
| A0   |  Eyebrow L pot wiper          -> PCA9685 ch 6
| A1   |  Eyebrow R pot wiper          -> PCA9685 ch 7
| A2   |  Basket Eyebrow L pot wiper   -> PCA9685 ch 8
| A3   |  Basket Eyebrow R pot wiper   -> PCA9685 ch 9
+------+

ADS_03 @ 0x4A (ADDR -> SDA)                 three face pots, right bank
+------+
| VDD  |  3.3V
| GND  |  GND
| SCL  |  TTGO GPIO 22 (I2C SCL)
| SDA  |  TTGO GPIO 21 (I2C SDA)
| ADDR |  SDA  = address 0x4A
| ALRT |  not connected
| A0   |  Nose pot wiper (up/down)     -> PCA9685 ch 10
| A1   |  FAULTY channel, nothing connected, left unread in firmware
| A2   |  Bottom Eyelid L pot wiper    -> PCA9685 ch 12
| A3   |  Bottom Eyelid R pot wiper    -> PCA9685 ch 13
+------+

ADS_04 @ 0x4B (ADDR -> SCL)                 neck-pivot + faders + Nose Basket
+------+
| VDD  |  3.3V
| GND  |  GND
| SCL  |  TTGO GPIO 22 (I2C SCL)
| SDA  |  TTGO GPIO 21 (I2C SDA)
| ADDR |  SCL  = address 0x4B
| ALRT |  not connected
| A0   |  NECK PIV pot -- DISABLED, read for display only (to be repurposed)
| A1   |  fader_left wiper             -> neck pivot, sole source (-2048..0..2048)
| A2   |  fader_right wiper            -> iris (doubles j4_display_right IRIS), 0-255
| A3   |  Nose Basket pot wiper        -> PCA9685 ch 11 (single source)
+------+
```

j4_receiver -- pin diagram

TTGO T-Display v1.1, rails visible, USB-C at the TOP (as mounted)

```
                          LEFT rail                                    RIGHT rail
                                            +--[ USB-C ]--+
                                 3.3V -- 3V3|             |  5V
                               ground -- GND|             |  GND  ground
                               ground -- GND|     TTGO    |  27   (free)
                               (free) -- 12 |  T-Display  |  26   (free)
                               (free) -- 13 |     v1.1    |  25   WS2812B data
                               (free) -- 15 |             |  33   I2C SCL (PCA9685)
    UART2 RX <- Teensy 24 (TX6) -- 2  |   [ ST7789    |  32   I2C SDA (PCA9685)
    UART2 TX -> Teensy 25 (RX6) -- 17 |    TFT on     |  39   (input only)
 UART1 RX <- j4_stepper_neck 13 -- 22 |    back ]     |  38   (input only)
  UART1 TX -> j4_stepper_neck 4 -- 21 |               |  37   (input only)
                               ground -- GND|             |  36   (input only)
                               ground -- GND|             |  3.3V
                                            +-------------+

    on-board (no header pin): GPIO34 = battery sense, BOOT = GPIO0,
        GPIO35 = screen mode button, TFT on 4/5/16/18/19/23
```

GPIO	Function
2	UART2 RX from Teensy (Teensy Serial6 TX, pin 24)
17	UART2 TX to Teensy (Teensy Serial6 RX, pin 25)
21	UART1 TX to j4_stepper_neck (250000 baud)
22	UART1 RX from j4_stepper_neck (status + EY: return)
25	WS2812B LED strip data (through ~330 ohm series resistor)
32	I2C SDA (PCA9685)
33	I2C SCL (PCA9685)
34	Battery voltage sense (ADC, read-only)
35	Screen mode button

j4_stepper_neck -- pin diagram

ESP32-D0WD (LOLIN32-Lite style), component side up, USB-C at the BOTTOM

```
        LEFT rail                                 RIGHT rail
                        +--------------+
 limit NR MIN -- 36 |   |              |   | 3V3   3.3V (limit pull-ups)
 limit NP MIN -- 39 |   |              |   | 22    DN UART2 TX -> eyes 4   (LED)
        reset -- EN |   |  ESP32-D0WD  |   | 19    limit NR MAX
 limit NL MIN -- 34 |   |  (LOLIN32-   |   | 23    limit NP MAX
DN RX <-eyes13 -- 35 |   |   Lite style)|   | 18    limit NL MAX
   DIR neck-R -- 32 |   |              |   | 5     (free)
  STEP neck-R -- 33 |   |              |   | 17    (free)
   DIR neck-L -- 25 |   |              |   | 16    (free)
  STEP neck-L -- 26 |   |              |   | 4     UART1 RX <- TTGO 21
  STEP neck-P -- 27 |   |              |   | 0     (free / BOOT)
   DIR neck-P -- 14 |   |              |   | 2     (free)
       (free) -- 12 |   |              |   | 15    (free)
       ground -- GND|   |              |   | 13    UART1 TX -> TTGO 22
                        +---[ USB-C ]--+

      SN74AHCT125N x2 (DIP-14, top view, notch at the top)

                      +---U---+
    1/OE (to GND) -- 1|       |14 -- VCC (5V)
    1A  (input)   -- 2|       |13 -- 4/OE (to GND)
    1Y  (output)  -- 3|       |12 -- 4A  (input)
    2/OE (to GND) -- 4|       |11 -- 4Y  (output)
    2A  (input)   -- 5|       |10 -- 3/OE (to GND)
    2Y  (output)  -- 6|       | 9 -- 3A  (input)
    GND           -- 7|       | 8 -- 3Y  (output)
                      +-------+

  U1: 1A<-26 1Y->NL PUL+ ; 2A<-25 2Y->NL DIR+ ; 3A<-33 3Y->NR PUL+ ; 4A<-32 4Y->NR DIR+
  U2: 1A<-27 1Y->NP PUL+ ; 2A<-14 2Y->NP DIR+ ; 3A,4A->GND (spare) ; 3Y,4Y n/c
  GPIO 21 is NOT present on this board, which is why the downstream TX is on 22.
  Neck STEP/DIR pins pass through the 74AHCT125 buffers (3.3V -> 5V) into the
  DM556TE PUL+/DIR+ inputs (PUL-/DIR- to GND). 100nF across each chip's VCC/GND;
  all /OE pins to GND.
  Limits: one NC switch per endpoint. MIN ends (34/36/39) need external 10k
  pull-ups to 3.3V; MAX ends (18/19/23) use the internal pull-ups.
```

GPIO	Function
26 / 25	neck-left STEP / DIR -> 74AHCT125 U1 -> DM556TE PUL+ / DIR+
33 / 32	neck-right STEP / DIR -> 74AHCT125 U1 -> DM556TE PUL+ / DIR+
27 / 14	neck-pivot STEP / DIR -> 74AHCT125 U2 -> DM556TE PUL+ / DIR+
4 / 13	Upstream UART1 RX / TX -- to j4_receiver: RX from TTGO GPIO 21, TX to TTGO GPIO 22
22 / 35	Downstream UART2 TX / RX -- to j4_stepper_eyes: TX to eyes GPIO 4, RX from eyes GPIO 13
34	neck-left MIN limit (input-only, external 10k pull-up)
36	neck-right MIN limit (input-only, external 10k pull-up)
39	neck-pivot MIN limit (input-only, external 10k pull-up)
18	neck-left MAX limit (internal pull-up)
19	neck-right MAX limit (internal pull-up)
23	neck-pivot MAX limit (internal pull-up)

j4_stepper_eyes -- pin diagram

ESP32-D0WD (LOLIN32-Lite style), component side up, USB-C at the BOTTOM

```
   LEFT rail                                RIGHT rail
                       +--------------+
limit ER MIN -- 36 |   |              |   | 3V3   3.3V (limit pull-ups)
limit EL MAX -- 39 |   |              |   | 22    (free / onboard LED)
       reset -- EN |   |  ESP32-D0WD  |   | 19    (free)
limit EL MIN -- 34 |   |  (LOLIN32-   |   | 23    (free)
limit ER MAX -- 35 |   |   Lite style)|   | 18    (free)
   DIR eye-R -- 32 |   |              |   | 5     (free)
  STEP eye-R -- 33 |   |              |   | 17    TMC2209 UART TX
   DIR eye-L -- 25 |   |              |   | 16    TMC2209 UART RX
  STEP eye-L -- 26 |   |              |   | 4     UART1 RX <- neck 22
      (free) -- 27 |   |              |   | 0     (free / BOOT)
      (free) -- 14 |   |              |   | 2     (free)
      (free) -- 12 |   |              |   | 15    (free)
      ground -- GND|   |              |   | 13    UART1 TX -> neck 35
                       +---[ USB-C ]--+

Both TMC2209 share the UART bus (17/16); addresses 0b00 (eye-L) / 0b01 (eye-R).
Limits: one NC switch per endpoint, each with a 10k pull-up to 3.3V
(input-only pins have no internal pull-up).
```

GPIO	Function
26 / 25	eye-left STEP / DIR
33 / 32	eye-right STEP / DIR
17 / 16	TMC2209 UART TX / RX (shared PDN_UART bus)
4 / 13	Upstream UART1 RX / TX (to neck board)
34	eye-left MIN limit (input-only, external 10k pull-up)
39	eye-left MAX limit (input-only, external 10k pull-up)
36	eye-right MIN limit (input-only, external 10k pull-up)
35	eye-right MAX limit (input-only, external 10k pull-up)

j4_talk -- pin diagram

Teensy 4.1 + Audio Shield Rev D, component side up, USB at the TOP

```
                          +======[ USB ]======+
                    GND  -| GND           5V |-  5V
                     0   -|  0           GND |-  GND
                     1   -|  1          3.3V |-  3.3V
front_LED_01 (T5)  2  *-|  2            23  |-  (audio I2S)
front_LED_02 (T4)  3  *-|  3            22  |-* front_LED_09 (T4)
front_LED_03 (T3)  4  *-|  4            21  |-  (audio I2S)
front_LED_04 (T2)  5  *-|  5            20  |-  (audio I2S)
front_LED_05 (T1)  6  *-|  6            19  |-  (audio I2C)
    (audio)        7   -|  7            18  |-  (audio I2C)
    (audio)        8   -|  8            17  |-* front_LED_08 (T3)
front_LED_06 (T1)  9  *-|  9            16  |-* front_LED_07 (T2)
 side_LED_07 (T5)  10 *-| 10            15  |-* side_LED_01 (T2)
 side_LED_08 (T5)  11 *-| 11            14  |-* front_LED_10 (T5)
                   12  -| 12            13  |-  (LED_BUILTIN)
 Serial6 TX        24 *-| 24            41  |-
 Serial6 RX        25 *-| 25            40  |-
                   26  -| 26            39  |-
                   27  -| 27            38  |-
 side_LED_02 (T2)  28 *-| 28            37  |-* side_LED_06 (T4)
 side_LED_03 (T3)  29 *-| 29            36  |-* side_LED_05 (T4)
                   30  -| 30            35  |-
                   31  -| 31            34  |-
                   32  -| 32         (bottom: SD card + SMD pads)
 side_LED_04 (T3)  33 *-| 33                 |
                          +===================+

 Serial6: 24 (TX) -> j4_receiver GPIO 2 ;  25 (RX) <- j4_receiver GPIO 17
 Audio Shield Rev D uses 7, 8, 20, 21, 23 (I2S) + 18, 19 (I2C) + built-in SD.
 front_LED_01..10 = front row left to right (pins 2,3,4,5,6,9,16,17,22,14);
 front 05/06 light first. side_LED_01..08 = down the side, top to bottom
 (pins 15,28,29,33,36,37,10,11).
```

Teensy Pin	Function
2	front_LED_01 (tier 5)
3	front_LED_02 (tier 4)
4	front_LED_03 (tier 3)
5	front_LED_04 (tier 2)
6	front_LED_05 (tier 1, innermost)
9	front_LED_06 (tier 1, innermost)
10	side_LED_07 (tier 5)
11	side_LED_08 (tier 5)
14	front_LED_10 (tier 5)
15	side_LED_01 (tier 2)
16	front_LED_07 (tier 2)
17	front_LED_08 (tier 3)
22	front_LED_09 (tier 4)
24	Serial6 TX to j4_receiver (receiver GPIO 2)
25	Serial6 RX from j4_receiver (receiver GPIO 17)
28	side_LED_02 (tier 2)
29	side_LED_03 (tier 3)
33	side_LED_04 (tier 3)
36	side_LED_05 (tier 4)
37	side_LED_06 (tier 4)

j4_display_left -- pin diagram

SeeedStudio XIAO ESP32S3, component side up, USB-C at the TOP

```
                LEFT rail                                      RIGHT rail
                               +--[ USB-C ]--+
       TFT BL  -- D0 (GP1) |  |             |  | 5V           (unused)
       TFT RST -- D1 (GP2) |  |    XIAO     |  | GND          ground
       TFT DC  -- D2 (GP3) |  |   ESP32-S3  |  | 3V3          TFT VCC
       TFT CS  -- D3 (GP4) |  |             |  | D10 (GP9)    TFT MOSI
        (free) -- D4 (GP5) |  |             |  | D9  (GP8)    TFT MISO
        (free) -- D5 (GP6) |  |             |  | D8  (GP7)    TFT SCLK
  LINK TX -> TTGO 27 -- D6 |  |             |  | D7  (GP44)   LINK RX <- TTGO 17
                (GP43)        +-------------+
```

XIAO Pin (GPIO)	Function
3V3 (n/a)	Display VCC (3.3V power)
GND (n/a)	Display GND, shared with TTGO GND
D0 (GPIO1)	TFT backlight (HIGH = on)
D1 (GPIO2)	TFT RST
D2 (GPIO3)	TFT DC
D3 (GPIO4)	TFT CS
D4 (GPIO5)	Free (I2C SDA if needed later)
D5 (GPIO6)	Free (I2C SCL if needed later)
D6 (GPIO43)	UART TX to TTGO GPIO27
D7 (GPIO44)	UART RX from TTGO GPIO17
D8 (GPIO7)	TFT SCLK
D9 (GPIO8)	TFT MISO
D10 (GPIO9)	TFT MOSI

j4_display_right -- pin diagram

SeeedStudio XIAO ESP32S3, component side up, USB-C at the TOP

```
           LEFT rail                               RIGHT rail
                            +--[ USB-C ]--+
      TFT BL  -- D0 (GP1) | |             | | 5V            (unused)
      TFT RST -- D1 (GP2) | |    XIAO     | | GND           ground
      TFT DC  -- D2 (GP3) | |   ESP32-S3  | | 3V3           TFT VCC + ADS1115 VDD
      TFT CS  -- D3 (GP4) | |             | | D10 (GP9)     TFT MOSI
     I2C SDA  -- D4 (GP5) | |             | | D9  (GP8)     TFT MISO
     I2C SCL  -- D5 (GP6) | |             | | D8  (GP7)     TFT SCLK
LINK TX -> TTGO 26 -- D6  | |             | | D7  (GP44)    LINK RX <- TTGO 25
             (GP43)         +-------------+

   D4/D5 are the XIAO's native I2C pins -- the ADS1115 hangs there (addr 0x48).
   The TTGO->XIAO direction (D7) carries the face-preset messages (M:/X:).
```

XIAO Pin (GPIO)	Function
3V3 (n/a)	Display VCC + ADS1115 VDD (3.3V power)
GND (n/a)	Display/ADS1115 GND, shared with TTGO GND
D0 (GPIO1)	TFT backlight (HIGH = on)
D1 (GPIO2)	TFT RST
D2 (GPIO3)	TFT DC
D3 (GPIO4)	TFT CS
D4 (GPIO5)	I2C SDA (ADS1115)
D5 (GPIO6)	I2C SCL (ADS1115)
D6 (GPIO43)	UART TX to TTGO GPIO26
D7 (GPIO44)	UART RX from TTGO GPIO25 (face-preset messages)
D8 (GPIO7)	TFT SCLK
D9 (GPIO8)	TFT MISO
D10 (GPIO9)	TFT MOSI

j4_display_right -- ADS1115 pin diagrams

pot wiper to the A-pin; outer legs to 3.3V and GND

```
ADS1115 @ 0x48 (ADDR -> GND)                on the XIAO's own I2C bus
+------+
| VDD  |  XIAO 3V3
| GND  |  XIAO GND
| SCL  |  XIAO D5 (GPIO 6)
| SDA  |  XIAO D4 (GPIO 5)
| ADDR |  GND  = address 0x48
| ALRT |  not connected
| A0   |  IRIS pot wiper        -> 270-deg iris servo (PCA9685 on j4_receiver)
| A1   |  COLOR pot wiper       -> WS2812B strip color (j4_receiver)
| A2   |  BRIGHTNESS pot wiper  -> WS2812B strip brightness (j4_receiver)
| A3   |  VOLUME pot wiper      -> j4_talk audio volume
+------+
```